20/12/2023
Regular

Qn. Set Code-4

Semester: 3rd
Programme: B.Tech
Branch: E&TC, E&EE, E&CSc

AUTUMN END SEMESTER EXAMINATION-2023

3rd Semester B.Tech

SIGNALS & SYSTEM

EC20001

(For 2023 (L.E), 2022 & Previous Admitted Batches)

Time: 3 Hours | Full Marks: 50

Answer any FIVE questions.
Question paper consists of two SECTIONS i.e. A and B.
SECTION-A is compulsory.
Attempt any TWO questions from SECTION-B.
The figures in the margin indicate full marks.
All parts of a question should be answered at one place only.

SECTION-A

1. Suppose that we want to design a discrete-time LTI system with the property that if input is $x(n) = \left(\frac{1}{2}\right)^n u(n) - \frac{1}{4}\left(\frac{1}{2}\right)^{n-1} u(n-1)$, then the output is $y(n) = \left(\frac{1}{3}\right)^n u(n)$. [10]

   a) Find the impulse response and frequency response of a discrete-time LTI system that has the foregoing property.

   b) Find the difference equation relating $x(n)$ and $y(n)$that characterizes the system.

OR

The input to a causal LTI system is $x[n] = u[-n-1] + \left(\frac{1}{2}\right)^n u[n]$. [10]

The z-transform of the output of this system is $Y(z) = \frac{-\frac{1}{2}z^{-1}}{\left(1-\frac{1}{2}z^{-1}\right)(1+z^{-1})}$.

   a) Determine $H(z)$, the z-transform of the system impulse response.

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b) What is the ROC for $Y(z)$.

c) Determine $y[n]$.

2. Demonstrate the property of convolution using 4-point DFT method, given $x_1(n) = \{1,2,3\}$, $x_2(n) = \{4,0,6\}$. [10]

OR

Apply overlap-add method to filter long data sequence $x(n) = \{1,2,3,4,5,6,7,8,9\}$ with the filter given by $h(n) = \{1,0,-1\}$. [10]

3. (a) Design an LPF with the following desired frequency response: [5]

$$H_d(e^{jw}) = \begin{cases} e^{-j3w} & \frac{-\pi}{4} \le w \le \frac{\pi}{4} \\ 0 & \frac{\pi}{4} < |w| \le \pi \end{cases}$$

Determine the filter coefficients $h(n)$ if the window function is defined using rectangular window.

(b) Determine the frequency response $H(e^{jw})$ of the designed filter. [5]

OR

(a) Design an HPF with the following desired frequency response: [5]

$$H_d(e^{jw}) = \begin{cases} 0 & \frac{-\pi}{4} \le w \le \frac{\pi}{4} \\ e^{-j3w} & \frac{\pi}{4} < |w| \le \pi \end{cases}$$

Determine the filter coefficients $h(n)$ if the window function is defined using rectangular window.

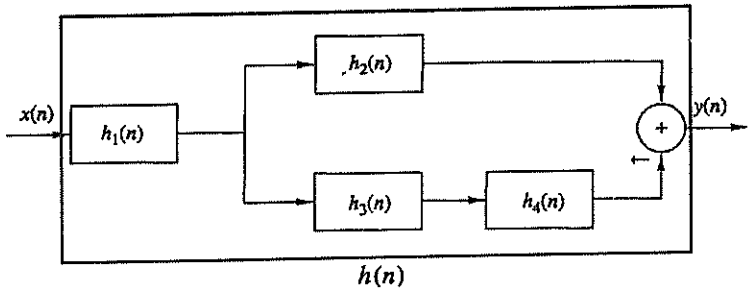
(b) Determine the frequency response $H(e^{jw})$ of the designed filter. [5]

SECTION-B

4. Build the Direct form-I and Direct form-II structure of the following difference equation [5+5]

$$-2y(n) - 6y(n-1) + 4y(n-2) = 6x(n-1) - 2x(n-2) + 4x(n-3).$$

5. (a) [5]



Consider the interconnected LTI system as shown in figure.

Determine overall $h(n)$, if $h_1(n) = \{1 \quad \underset{\uparrow}{2} \quad 1\}$,

$h_2(n) = \{3 \quad \underset{\uparrow}{2} \quad 1\}$, $h_3(n) = \{1 \quad 1\}$,

$h_4(n) = \{2 \quad 2\}$ are given respectively.

(b) Determine the response of the above system, if the input is $x(n) = \delta(n+1) + \delta(n) - \delta(n-1)$. [5]

6. The system function of a causal LTI system is $H(z) = \frac{1-z^{-1}}{1+\frac{3}{4}z^{-1}}$. The input to this system is $x[n] = \left(\frac{1}{3}\right)^n u[n] + u[-n-1]$. [10]

a) Find the impulse response of the system $h[n]$.

b) Find the output $y[n]$.

c) Is the system stable?

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